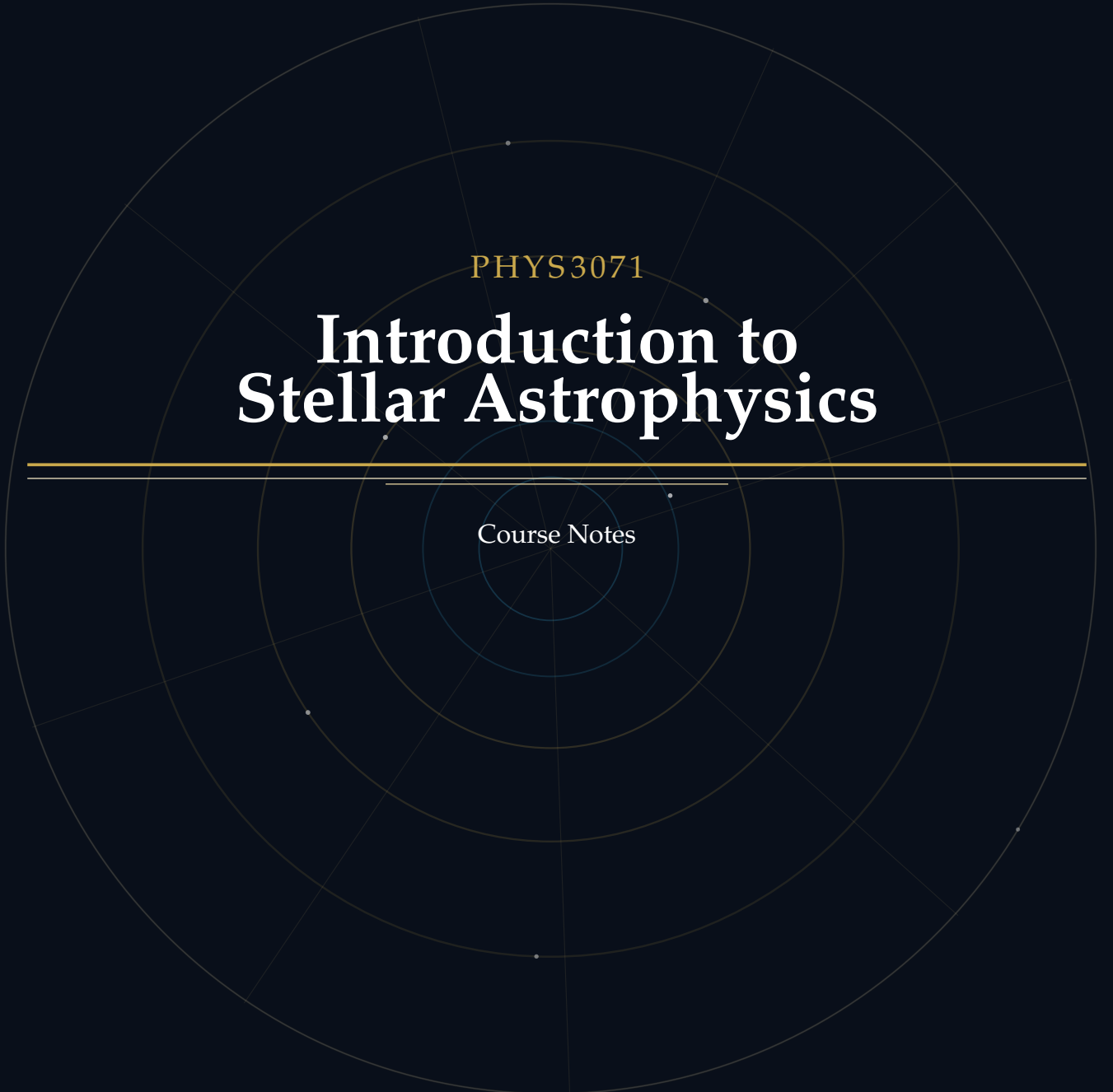




PHYS3071

Introduction to Stellar Astrophysics

Course Notes

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Introduction

About These Notes

These notes accompany the PHYS3071 Introduction to Stellar Astrophysics course in Hong Kong University of Science and Technology (HKUST). They are intended as a self-contained supplement to the lectures, providing both conceptual grounding and mathematical rigour at a level appropriate for readers with a background in special relativity, basic linear algebra, and basic calculus.

The material focuses on core stellar-astrophysics topics, including stellar structure, energy generation in stars, radiative transport, and observational diagnostics such as the Hertzsprung–Russell diagram. Each topic is structured to serve different learning objectives—from building physical intuition to solving examination-style problems and exploring advanced derivations.

These notes do not replace lecture slides or the recommended textbooks. They are best used alongside them as a structured study companion.

If you find any errors or have suggestions for improvement, please feel free to contact me at yhip@connect.ust.hk.

Document Conventions

Block Components

Each topic is presented using a consistent **5-component block structure** designed to support multiple learning styles:

PHYSICS INTUITION

Purpose: Builds conceptual understanding through plain language explanations.

This section provides the *motivation* behind the physics concept, explores its *historical development*, and offers *intuitive explanations* that help you grasp what the theory is really about before diving into equations.

Use this when: You're first encountering a new concept or need to refresh the "big picture" before tackling mathematics.

THEORY

Purpose: Provides rigorous mathematical treatment of the concept. Here you'll find detailed *derivations*, *fundamental assumptions*, *key equations*, and connections between *mathematics and physical intuition*. This is the technical foundation required for exams and deeper problem-solving.

Use this when: You need to understand how to apply formulas, prepare for exams, or work through problems rigorously.

QUESTION EXAMPLES

Purpose: Demonstrates practical application through worked problems.

Each question example mimics exam-style problems and includes complete *solution walkthroughs*. These help you bridge the gap between theory and problem-solving, and expose common pitfalls.

Use this when: Practicing problem-solving, preparing for assessments, or checking if you can apply the formulas correctly.

FOUNDATION

Purpose: Supplies background knowledge and detailed proof steps that underpin the main content.

This section presents *prerequisite mathematics or physics*, *step-by-step derivations* that are too detailed to include in the Theory block, and *intermediate results* needed for a thorough understanding. Reading this is optional for the exam but essential for genuine mastery.

Use this when: A derivation in Theory feels too abrupt, you want to verify an intermediate step, or you need to solidify prerequisite concepts.

EXTRA

Purpose: Provides supplementary material that extends *beyond* the course syllabus.

This section offers *interesting asides*, *historical context*, *connections to current research*, and citations to deeper references. The content here is intentionally out-of-scope for the course examination but enriches the broader picture of each topic.

Use this when: You're curious to explore beyond the syllabus, or when a topic connects to your personal research interests.

Suggested Reading Paths:

1. **Exam Preparation:** *Physics Intuition* → *Theory* → *Question Examples*
2. **Deep Learning:** *Physics Intuition* → *Foundation* → *Theory* → *Question Examples* → *Extra*
3. **Quick Review:** *Theory* → *Question Examples*

Special Annotations

In addition to the five block components above, two inline annotations appear throughout the text to flag specific types of information:

IMPORTANT NOTE

Purpose: Alerts you to a result, condition, or sign convention that is easy to get wrong or is frequently tested.

An Important Note signals a *common misconception*, a *critical sign*, a *non-obvious condition* (e.g. a constraint that must hold for an equation to be valid), or an *exam trap* worth memorising explicitly.

Treat these as mandatory reading regardless of which reading path you follow.

REMARK

Purpose: Highlights a useful observation, connection, or naming convention.

A Remark draws attention to a *conceptual link* between results, an *elegant reformulation*, a *naming convention* (e.g. why a quantity carries its particular name), or a *minor but clarifying point* that does not rise to the level of an Important Note.

Use this when: Looking for a deeper understanding of why an equation takes the form it does, or for useful mnemonics.

Mathematical Conventions

The following conventions are used throughout these notes:

Metric signature. When relativistic expressions are needed, we adopt the *mostly-plus* (or “ $(-, +, +, +)$ ”) signature:

$$g_{\mu\nu} = \text{diag}(-1, +1, +1, +1).$$

IMPORTANT NOTE

Many general-relativity textbooks use the opposite, *mostly-minus* $(+, -, -, -)$ signature. The two conventions differ by an overall sign in the Ricci scalar and stress-energy trace. Always check the signature convention when comparing formulae across sources.

Natural units. In sections involving high-energy or compact-object estimates, we may set $c = \hbar = k_B = 1$. Quantities are then expressed in powers of eV (or GeV). Explicit c and $\hbar$ factors are restored when numerical values are quoted in SI units.

Key symbols.

Symbol	Meaning	Typical units
$M_{\odot}$	Solar mass	kg
$R_{\odot}$	Solar radius	m
$L_{\odot}$	Solar luminosity	W
T_{eff}	Effective temperature	K
μ	Mean molecular weight	—
ρ	Mass density	kg m^{-3}

REMARK

Unless otherwise stated, stellar quantities are quoted in solar-normalised form (for example $M/M_{\odot}$ and $L/L_{\odot}$), which makes scaling relations easier to compare across different stellar masses.

differential notation. We use $\dot{x} \equiv \frac{dx}{dt}$ for time derivatives of some variables $x(t)$.

1.1 Observational Foundations

1.1.1 Specific Intensity, Flux, and Luminosity

PHYSICS INTUITION

Observed starlight is measured in layers: direction-resolved intensity, surface flux, and finally total emitted power. Keeping these definitions separate avoids unit mistakes in later derivations.

THEORY

- Specific intensity I_ν :

$$dE = I_\nu \cos \theta dA dt d\nu d\Omega \quad (1.1)$$

- Monochromatic flux density:

$$F_\nu = \int I_\nu \cos \theta d\Omega \quad (1.2)$$

- Bolometric flux:

$$F = \int_0^\infty F_\nu d\nu \quad (1.3)$$

- Luminosity:

$$L = \int F dA \quad (1.4)$$

IMPORTANT NOTE

I_ν is invariant along a ray in vacuum (no absorption/emission), while F_ν depends on source geometry and distance. Do not interchange them in magnitude or distance calculations.

1.1.2 Magnitude System and Distance Modulus

THEORY

$$M_1 - M_2 = -2.5 \log_{10} \left(\frac{L_1}{L_2} \right), \quad (1.5)$$

$$m_1 - m_2 = -2.5 \log_{10} \left(\frac{F_1}{F_2} \right), \quad (1.6)$$

$$m - M = 5 \log_{10}(d/\text{pc}) - 5, \quad (1.7)$$

$$F = \frac{L}{4\pi d^2}. \quad (1.8)$$

Question Example 1.1: Distance modulus quick use

A star has $m = 9.0$ and $M = 4.0$. Estimate its distance.

Solution.

$$\begin{aligned} m - M &= 5 \log_{10}(d/\text{pc}) - 5 = 5 \\ \Rightarrow \log_{10}(d/\text{pc}) &= 2 \\ \Rightarrow d &= 10^2 \text{ pc} = 100 \text{ pc}. \end{aligned}$$

1.1.3 Parallax and Radius

THEORY

- Parallax distance:

$$d(\text{pc}) = \frac{1}{\pi(\text{arcsec})} \quad (1.9)$$

- Radius from angular diameter θ :

$$R = \frac{1}{2} \theta d \quad (1.10)$$

1.2 Basic Stellar Parameters

1.2.1 Surface Temperature and Blackbody Radiation

THEORY

- Planck function:

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1} \quad (1.11)$$

- Stefan–Boltzmann law:

$$F = \sigma T^4, \quad \sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4} \quad (1.12)$$

- Luminosity:

$$L = 4\pi R^2 \sigma T_{\text{eff}}^4 \quad (1.13)$$

- Wien’s law:

$$\lambda_{\text{max}} T = 2.9 \times 10^{-3} \text{ m K} \quad (1.14)$$

1.2.2 Mass and Main-Sequence Lifetime

THEORY

- Hydrogen-burning mass window is roughly

$$0.08 M_\odot \lesssim M \lesssim 100 M_\odot. \quad (1.15)$$

- Binary stars (Kepler’s third law):

$$\frac{a^3}{P^2} = \frac{G(M_1 + M_2)}{4\pi^2} \quad (1.16)$$

- Main-sequence scaling:

$$\tau \propto \frac{M}{L} \propto M^{-2.5} \quad (1.17)$$

REMARK

Low-mass stars are dim but long-lived; high-mass stars are bright but short-lived. This anti-correlation is central to cluster age dating.

1.3 From Interstellar Clouds to Protostars

1.3.1 Nebulae and Collapse Triggers

PHYSICS INTUITION

Star formation starts in cold molecular gas where gravity can eventually beat thermal and magnetic support. External perturbations often provide the last

push.

THEORY

- Emission nebulae: ionized gas emits line radiation.
- Reflection nebulae: dust scatters short wavelengths more efficiently ($\propto \lambda^{-4}$).
- Dark nebulae: dust extinction blocks background light.

Typical collapse triggers include supernova shocks, cloud–cloud collisions, and compression by nearby young stars.

1.3.2 Simplified Cloud Model and Hydrostatic Balance

FOUNDATION

Assume spherical symmetry and radial dependence only. Neglect rotation, magnetic fields, and turbulence in the first model to isolate gravity-versus-pressure physics.

THEORY

$$dm(r, t) = 4\pi r^2 \rho dr, \quad (1.18)$$

$$\frac{\partial m}{\partial t} + v \frac{\partial m}{\partial r} = 0, \quad (1.19)$$

$$\frac{dP}{dr} = -\frac{Gm(r)\rho}{r^2} = -\rho g. \quad (1.20)$$

1.4 Virial Theorem and Jeans Criterion

1.4.1 Energetics of Self-Gravitating Gas

THEORY

$$\langle P \rangle = -\frac{1}{3} \frac{E_{\text{GR}}}{V}, \quad (1.21)$$

$$E_{\text{GR}} = -\int_0^M \frac{Gm}{r} dm = -f_{\text{GR}} \frac{GM^2}{R}, \quad (1.22)$$

$$P = \frac{1}{3} nm \langle v^2 \rangle = \frac{2}{3} n \langle E_k \rangle. \quad (1.23)$$

For uniform density, $f_{\text{GR}} \approx 3/5$.

1.4.2 Jeans Instability

THEORY

$$|E_{\text{GR}}| > 2E_{\text{th}} \quad (\text{collapse}), \quad (1.24)$$

$$M_J = \left(\frac{5k_B T}{G\mu m_H} \right)^{3/2} \left(\frac{3}{4\pi\rho} \right)^{1/2}, \quad (1.25)$$

$$\lambda_J = \sqrt{\frac{\pi c_s^2}{G\rho}}, \quad (1.26)$$

$$M_J \propto T^{3/2} \rho^{-1/2}. \quad (1.27)$$

IMPORTANT NOTE

Lower temperature and higher density both reduce M_J . This is why cooling strongly promotes fragmentation and cluster formation.

1.4.3 Fragmentation and End of Contraction

EXTRA

Massive collapsing clouds fragment when local regions exceed their own Jeans threshold, producing stellar groups rather than a single object. As opacity rises and ionization increases, trapped radiation raises pressure and leads to a protostar. Contraction ends either by ignition of hydrogen burning ($T \gtrsim 10^7$ K) or by degeneracy support (brown-dwarf pathway).

2.1 First Law and Heat Capacities

2.1.1 Fundamental Identity

THEORY

For systems with variable particle number,

$$dQ = dU + dW - \mu dN, \quad (2.1)$$

with $dW = P dV$. For closed systems ($dN = 0$),

$$dQ = dU + P dV. \quad (2.2)$$

IMPORTANT NOTE

Sign conventions matter. In these notes, $dQ > 0$ means heat into the system and $dW = P dV$ is work done by the system.

2.1.2 Adiabatic, Isochoric, Isobaric, Isothermal Processes

THEORY

- Adiabatic ($dQ = 0$):

$$dU = -P dV \quad (2.3)$$

- Isochoric ($dV = 0$):

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V \quad (2.4)$$

- Isobaric ($P = \text{const}$, ideal gas):

$$C_P = C_V + nR \quad (2.5)$$

- Isothermal (ideal gas):

$$dU = 0, \quad dQ = dW = \frac{nRT}{V} dV \quad (2.6)$$

2.2 Entropy and Thermodynamic Potentials

2.2.1 Entropy and Fundamental Relation

THEORY

For reversible processes,

$$dS = \frac{dQ_{\text{rev}}}{T}, \quad (2.7)$$

leading to

$$dU = T dS - P dV + \mu dN. \quad (2.8)$$

2.2.2 Helmholtz Free Energy and Maxwell Relations

FOUNDATION

Define $F = U - TS$. Then

$$dF = -S dT - P dV + \mu dN, \quad (2.9)$$

which gives, for example,

$$\left(\frac{\partial S}{\partial V}\right)_{T,N} = \left(\frac{\partial P}{\partial T}\right)_{V,N}. \quad (2.10)$$

REMARK

For ideal gas, $(\partial U/\partial V)_T = 0$, so internal energy depends only on T . This fails for interacting gases such as Van der Waals fluids.

2.3 Photon Gas and Blackbody Radiation

2.3.1 Photon Statistics and EOS

THEORY

Photons are bosons with $\mu = 0$ and $\varepsilon = pc$. With Bose–Einstein statistics,

$$\langle n_p \rangle = \frac{1}{e^{pc/k_B T} - 1}. \quad (2.11)$$

The energy density is

$$u = aT^4, \quad a = \frac{8\pi^5 k_B^4}{15h^3 c^3}, \quad (2.12)$$

and the pressure is

$$P_{\text{rad}} = \frac{1}{3}u = \frac{1}{3}aT^4. \quad (2.13)$$

2.3.2 Adiabatic Expansion of Radiation

THEORY

For photon gas with $dQ = 0$,

$$T^3V = \text{const}, \quad PV^{4/3} = \text{const}. \quad (2.14)$$

So the adiabatic index is $\gamma = 4/3$.

2.3.3 Planck Spectrum and Stefan–Boltzmann Law

THEORY

$$u(\nu)d\nu = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1} d\nu, \quad (2.15)$$

$$I_\nu = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1}, \quad (2.16)$$

$$F = \sigma T^4, \quad \sigma = \frac{ac}{4}. \quad (2.17)$$

2.4 Thermodynamics in Stellar Interiors

2.4.1 Gas and Radiation Pressure

THEORY

In a fully ionized plasma,

$$P_{\text{total}} = \frac{\rho k_B T}{\mu m_u} + \frac{1}{3}aT^4. \quad (2.18)$$

Radiation pressure becomes important in massive stars.

2.4.2 Energy Transport Overview

PHYSICS INTUITION

Stellar heat transport transitions between conduction, radiation, and convection depending on opacity and stability of stratification. The dominant mechanism is a structural diagnostic for stellar mass and evolutionary stage.

THEORY

- Mean free path: $\lambda = 1/(n\sigma)$.
- Conductive flux: $j = -\kappa dT/dx$.
- Radiative diffusion:

$$F_{\text{rad}} = -\frac{4ac}{3\kappa\rho} T^3 \frac{dT}{dr}. \quad (2.19)$$

- Convection criterion (Schwarzschild):

$$\nabla \equiv \frac{d \ln T}{d \ln P} > \nabla_{\text{ad}}. \quad (2.20)$$

EXTRA

Mass dependence of transport zones (fully convective low-mass stars, solar-type radiative core + convective envelope, massive-star convective cores) can be understood from opacity trends and temperature sensitivity of nuclear burning.

2.5 Eddington Luminosity

THEORY

Hydrostatic support requires radiative acceleration not to exceed gravity:

$$L < L_{\text{Edd}} \equiv \frac{4\pi GMc}{\kappa_R}. \quad (2.21)$$

For electron scattering opacity,

$$L_{\text{Edd}} \approx 1.3 \times 10^{31} \text{ W} \left(\frac{M}{M_{\odot}} \right). \quad (2.22)$$

3.1 Characteristic Stellar Temperatures

3.1.1 Internal Temperature Scale

THEORY

Using hydrostatic and virial scaling for an ideal-gas dominated star,

$$k_B T \sim \frac{1}{3} \frac{GM\mu m_u}{R}. \quad (3.1)$$

Typical stellar cores satisfy $T_I \sim 10^6$ – 10^7 K.

3.1.2 Effective Surface Temperature

THEORY

From Stefan–Boltzmann law,

$$L = 4\pi R^2 \sigma T_E^4, \quad T_E = \left(\frac{L}{4\pi R^2 \sigma} \right)^{1/4}, \quad (3.2)$$

with typical values $T_E \sim 10^3$ – 10^4 K.

REMARK

The large gap between T_I and T_E is a direct consequence of optical depth: radiation random-walks through dense stellar matter before escaping.

3.2 Validity of Ideal-Gas Approximation

3.2.1 Inter-particle Distance and Coulomb Scale

FOUNDATION

Average spacing is

$$d \approx \left(\frac{\mu m_u}{\rho} \right)^{1/3}. \quad (3.3)$$

A rough Coulomb scale is

$$E_C \sim \frac{Z^2 e^2}{4\pi\epsilon_0 d}. \quad (3.4)$$

When thermal energy exceeds this scale, ideal-gas treatment is often sufficient.

3.2.2 Virial Estimate for Thermal Energy**THEORY**

For a self-gravitating system,

$$2E_{\text{kin}} + E_{\text{grav}} = 0, \quad (3.5)$$

and for ideal gas,

$$E_{\text{kin}} = \frac{3}{2} \frac{M}{\mu m_u} k_B T. \quad (3.6)$$

Combined with $E_{\text{grav}} \sim -GM^2/R$ gives the standard central-temperature estimate.

3.3 Classical and Relativistic Ideal Gases**3.3.1 Energy–Pressure Relations****THEORY**

General kinetic form:

$$P = \frac{1}{3} n \langle \vec{p} \cdot \vec{v} \rangle. \quad (3.7)$$

Hence

$$P = \frac{2}{3} u \quad (\text{NR}), \quad (3.8)$$

$$P = \frac{1}{3} u \quad (\text{UR}). \quad (3.9)$$

3.3.2 Maxwell–Boltzmann Limit

THEORY

For classical non-relativistic particles,

$$f(p) = A \exp \left(-\frac{p^2}{2mk_B T} \right), \quad (3.10)$$

which leads to the ideal-gas law:

$$PV = Nk_B T. \quad (3.11)$$

The same equation of state also holds for ultra-relativistic classical gas, though with different energy scaling.

3.4 Quantum Statistics and Degeneracy

3.4.1 Fermi–Dirac and Bose–Einstein Distributions

THEORY

$$f_{\text{FD}}(\epsilon) = \frac{1}{e^{(\epsilon-\mu)/k_B T} + 1}, \quad (3.12)$$

$$f_{\text{BE}}(\epsilon) = \frac{1}{e^{(\epsilon-\mu)/k_B T} - 1}. \quad (3.13)$$

Classical limit occurs when $e^{(\epsilon-\mu)/k_B T} \gg 1$.

3.4.2 Quantum Criterion

THEORY

Define quantum concentration

$$n_Q = g_s \left(\frac{2\pi mk_B T}{h^2} \right)^{3/2}. \quad (3.14)$$

Then:

- Classical regime: $n \ll n_Q$
- Quantum regime: $n \gtrsim n_Q$

IMPORTANT NOTE

Degenerate pressure can dominate even when temperature is low. It is set primarily by density through phase-space filling, not by thermal agitation.

3.4.3 Degenerate Fermi Gas at $T \rightarrow 0$ **THEORY**

At $T = 0$, states are filled up to p_F :

$$p_F = \hbar(3\pi^2 n)^{1/3} \quad (g_s = 2). \quad (3.15)$$

Degenerate pressure scales as

$$P_{\text{NR,deg}} \propto n^{5/3} \propto \rho^{5/3}, \quad (3.16)$$

$$P_{\text{UR,deg}} \propto n^{4/3} \propto \rho^{4/3}. \quad (3.17)$$

Question Example 3.1: Why white dwarfs shrink as mass increases

In the non-relativistic degenerate regime, equilibrium between gravity and $P \propto \rho^{5/3}$ leads to the approximate mass–radius trend $R \propto M^{-1/3}$.

Interpretation. Adding mass increases gravity faster than degeneracy support can maintain the same size, so the star compresses.

EXTRA

The transition from $\rho^{5/3}$ to $\rho^{4/3}$ support explains why there is a maximum white-dwarf mass (Chandrasekhar limit).

4.1 Nuclear Basics

4.1.1 Nuclear Size and Binding Energy

THEORY

$$R = R_0 A^{1/3}, \quad R_0 \approx 1.2 \times 10^{-15} \text{ m}, \quad (4.1)$$

$$BE(Z, N) = [Zm_p + Nm_n - M(Z, N)] c^2. \quad (4.2)$$

Binding energy per nucleon increases for light nuclei and decreases for very heavy nuclei due to Coulomb repulsion.

4.1.2 Why Fusion Powers Stars

PHYSICS INTUITION

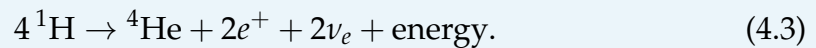
Stars gain energy when fusing light nuclei toward higher binding energy per nucleon. This trend reverses near iron, where further fusion is endothermic.

4.2 Hydrogen and Helium Burning

4.2.1 Proton–Proton Chain

THEORY

Representative net reaction:



Total released energy is about 26.7 MeV, with part carried away by neutrinos.

4.2.2 CNO Cycle

THEORY

CNO uses C, N, O as catalysts for H-burning and is much more temperature sensitive than pp burning. Near solar-core temperatures, pp-chain is weaker in T -sensitivity ($\sim T^4$) while CNO can scale roughly as $T^{\sim 18}$.

4.2.3 Triple-Alpha Process**THEORY**

enabled by resonance through unstable ${}^8\text{Be}$ and an excited Hoyle state in ${}^{12}\text{C}$. Typical ignition requires $T \gtrsim 10^8$ K.

IMPORTANT NOTE

The first pp step ($p + p \rightarrow d + e^+ + \nu_e$) is a weak-interaction bottleneck. It is slow enough to set long stellar main-sequence lifetimes.

4.3 Barrier Penetration and Reaction Rates**4.3.1 Coulomb Barrier and Tunneling****FOUNDATION**

Classically, nuclei with $E < V_C$ cannot fuse. Quantum tunneling gives non-zero penetration probability,

$$P \sim \exp \left[-\sqrt{\frac{E_G}{E}} \right], \quad (4.5)$$

where E_G is the Gamow energy.

4.3.2 Thermally Averaged Cross Section

THEORY

$$\langle \sigma v \rangle = \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(k_B T)^{3/2}} \int_0^\infty S(E) \exp \left[-\frac{E}{k_B T} - \sqrt{\frac{E_G}{E}} \right] dE. \quad (4.6)$$

The integrand peaks at the Gamow energy window, balancing high-energy tail abundance and tunneling probability.

4.3.3 Temperature Sensitivity**THEORY**

Near a reference temperature, reaction rates are often approximated by

$$R \propto T^\nu, \quad (4.7)$$

with small ν for pp-chain and much larger ν for CNO and triple-alpha. This sensitivity drives core structure differences between low-mass and massive stars.

Question Example 4.1: Qualitative core response to heating

Suppose core temperature rises slightly.

Discussion. In a pp-dominated core, energy generation increases moderately, giving mild feedback. In a CNO-dominated core, generation rises steeply, making burning more centrally concentrated and favoring convective cores.

4.4 Advanced Burning and Stellar Endpoints**4.4.1 Late Burning Stages in Massive Stars****THEORY**

After He burning, successive C, Ne, O, and Si burning stages occur at progressively higher temperatures and shorter timescales. Fusion energy production terminates near iron-group nuclei.

4.4.2 Neutrino Cooling and Rapid Evolution

EXTRA

At high temperature, neutrino processes (e.g., pair production) carry energy away efficiently. This accelerates late-stage evolution and contributes to core collapse in massive stars.

4.5 Solar Neutrino Problem

THEORY

Historically, measured solar neutrino flux was below early predictions. The resolution is flavor oscillation:

$$\nu_e \leftrightarrow \nu_\mu \leftrightarrow \nu_\tau, \quad (4.8)$$

so detectors sensitive mostly to ν_e observed an apparent deficit.

REMARK

Solar neutrino measurements are both a test of stellar fusion models and direct evidence for neutrino mass and mixing.

5.1 Core Stellar Structure Equations

5.1.1 Four Coupled ODEs

THEORY

For spherical stars:

$$\frac{dP}{dr} = -\frac{Gm\rho}{r^2}, \quad (5.1)$$

$$\frac{dm}{dr} = 4\pi r^2 \rho, \quad (5.2)$$

$$\frac{dT}{dr} = -\frac{3\kappa\rho}{16\pi acT^3} \frac{L(r)}{r^2}, \quad (5.3)$$

$$\frac{dL}{dr} = 4\pi r^2 \rho \varepsilon. \quad (5.4)$$

Boundary conditions include $m(0) = 0$, $L(0) = 0$, and photospheric constraints near $r = R$.

5.1.2 Closure Relations

FOUNDATION

The structure equations require constitutive inputs:

- Equation of state: $P = P(\rho, T)$
- Opacity: $\kappa = \kappa(\rho, T, \text{composition})$
- Nuclear source term: $\varepsilon = \varepsilon(\rho, T, \text{composition})$

5.2 Equations of State and Opacity

5.2.1 Gas, Radiation, and Degeneracy Pressure

THEORY

$$P_{\text{gas}} = \frac{\rho k_B T}{\mu m_H}, \quad (5.5)$$

$$P_{\text{rad}} = \frac{1}{3} a T^4, \quad (5.6)$$

$$P_{\text{NR,deg}} = K_{\text{NR}} \rho^{5/3}, \quad (5.7)$$

$$P_{\text{UR,deg}} = K_{\text{UR}} \rho^{4/3}. \quad (5.8)$$

Total pressure is the appropriate sum for the relevant regime.

5.2.2 Mean Molecular Weight**THEORY**

For ionized composition fractions (X, Y, Z) ,

$$\frac{1}{\mu} \approx 2X + \frac{3}{4}Y + \frac{1}{2}Z. \quad (5.9)$$

This parameter maps composition directly into the gas-pressure term.

5.2.3 Opacity and Nuclear Heating Scalings**THEORY**

Useful approximations:

$$\kappa_{\text{es}} \approx 0.02(1 + X) \text{ m}^2 \text{ kg}^{-1}, \quad (5.10)$$

$$\kappa_{\text{bf+ff}} \propto \rho T^{-3.5}, \quad (5.11)$$

$$\varepsilon_{pp} \propto \rho X^2 T^4. \quad (5.12)$$

REMARK

Opacity controls transport efficiency, while $\varepsilon(T)$ controls how centrally concentrated energy generation is. Together they determine whether stellar zones are radiative or convective.

5.3 Analytic Scaling Models**5.3.1 Parameterized Density Profiles**

PHYSICS INTUITION

Exact stellar models are numerical, but simple analytic profiles are valuable for quick estimates and physical intuition. They expose which observables mostly constrain central density, pressure, and temperature.

THEORY

A representative profile is

$$\rho(r) = \rho_c \exp(-r^2/a^2) \quad (5.13)$$

or polynomial alternatives. With hydrostatic balance and ideal-gas EOS, one derives approximate $P(r)$ and $T(r)$ trends from the center to the surface.

5.4 Stellar Mass Limits

5.4.1 Minimum Mass for Hydrogen Burning

THEORY

Competition between thermal support and electron degeneracy yields a minimum mass:

$$M_{\min} \approx 0.08 M_{\odot}. \quad (5.14)$$

Below this threshold, sustained H fusion does not ignite (brown dwarf regime).

5.4.2 Upper Mass and Radiation Support

THEORY

As mass increases, radiation pressure becomes significant, and instability / strong winds limit long-term growth. Observed and theoretical upper masses are typically of order 100–150 $M_{\odot}$.

IMPORTANT NOTE

The lower mass limit is set by degeneracy suppressing core heating; the upper limit is set by radiation-driven instability and mass loss. They arise from different physics and should not be conflated.

5.5 Compact Remnants

5.5.1 White Dwarfs

THEORY

White dwarfs are supported by electron degeneracy pressure, with characteristic behavior:

$$R \propto M^{-1/3} \quad (\text{NR degenerate limit}), \quad (5.15)$$

and maximum mass near

$$M_{\text{Ch}} \approx 1.44 \left(\frac{2}{\mu_e} \right)^2 M_{\odot}. \quad (5.16)$$

5.5.2 Cooling of White Dwarfs

THEORY

With ions as main heat reservoir and radiative surface losses,

$$L \propto t^{-7/5} \quad (5.17)$$

in simplified cooling models (modified by crystallization and envelope physics).

5.5.3 Neutron Stars (Overview)

EXTRA

Neutron stars are relativistic compact objects requiring TOV hydrostatic balance:

$$\frac{dP}{dr} = - \frac{G[\rho + P/c^2][m + 4\pi r^3 P/c^2]}{r^2 [1 - 2Gm/(rc^2)]}. \quad (5.18)$$

Observed masses around $2 M_{\odot}$ strongly constrain the high-density EOS.

Question Example 5.1: Regime check from mass scale

Compare two stars qualitatively:

- $0.05 M_{\odot}$: below H-burning threshold, degeneracy appears early.
- $30 M_{\odot}$: strong radiation pressure and high CNO sensitivity.

Conclusion. Low-mass objects trend toward brown-dwarf behavior, while high-mass stars show luminous, unstable, short-lived evolution.

A.1 Use of Artificial Intelligence in This Document

This course guide was developed with assistance from artificial intelligence (AI) in the following capacities:

- **Template Design:** AI provided guidance on structuring the document layout, choosing appropriate color schemes.
- **LaTeX Code Generation:** AI assisted in writing and refactoring LaTeX code, including package configurations, box styling (tcolorbox), TikZ decorations, and custom command definitions.
- **LaTeX Structure Refactoring:** AI helped reorganize and optimize the document structure for maintainability, readability, and professional presentation.
- **Mathematical Formula Formatting:** AI provided assistance in formatting and presenting mathematical equations in clear, readable notation aligned with conventions.

Author Verification: All content, including scientific accuracy, theoretical derivations, example problems, and pedagogical choices, has been carefully reviewed and verified by the author. The AI assistance was limited to technical formatting and structural optimization, not to the generation or validation of course content itself.

B.1 Useful Identities

Trigonometric

$$\sin^2 \theta + \cos^2 \theta = 1, \quad \tan \theta = \sin \theta / \cos \theta, \quad \sin(A \pm B) = \sin A \cos B \pm \cos A \sin B.$$

Hyperbolic

$$\cosh^2 x - \sinh^2 x = 1, \quad \sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y.$$

Common Integrals

$$\begin{aligned} \int x^n dx &= \frac{x^{n+1}}{n+1} + C \quad (n \neq -1), & \int e^{ax} dx &= \frac{1}{a} e^{ax} + C, \\ \int \frac{dx}{x} &= \ln |x| + C, & \int \sin x dx &= -\cos x + C. \end{aligned}$$

Taylor Expansions

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \dots$$

$$\sqrt{1+x} = \sum_{n=0}^{\infty} \binom{1/2}{n} x^n = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \dots$$

B.2 Differential Geometry Primer

The **metric tensor** $g_{\mu\nu}$ encodes the geometry of spacetime: $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$. The **Christoffel symbols** (not tensors) are defined as

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}). \quad (\text{B.1})$$

The Riemann curvature tensor is defined as

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma_{\nu\sigma}^\rho - \partial_\nu \Gamma_{\mu\sigma}^\rho + \Gamma_{\mu\lambda}^\rho \Gamma_{\nu\sigma}^\lambda - \Gamma_{\nu\lambda}^\rho \Gamma_{\mu\sigma}^\lambda. \quad (\text{B.2})$$

The **Ricci tensor** is the contraction $R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu}$ of the Riemann curvature tensor, and the **Ricci scalar** is $R = g^{\mu\nu} R_{\mu\nu}$.

These enter the Einstein field equations as:

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}. \quad (\text{B.3})$$

Mostly-Plus vs. Mostly-Minus Signatures

The choice of metric signature affects the signs of above tensors in below ways:

Tensor	Symbol	Sign Change
Metric	$g_{\mu\nu}$	Flips (−)
Inverse metric	$g^{\mu\nu}$	Flips (−)
Christoffel symbols	$\Gamma_{\mu\nu}^\lambda$	No change
Riemann(1,3)	$R^\rho{}_{\sigma\mu\nu}$	No change
Riemann(0,4)	$R_{\rho\sigma\mu\nu}$	Flips (−)
Ricci tensor	$R_{\mu\nu}$	No change
Ricci scalar	R	Flips (−)
Einstein tensor	$G_{\mu\nu}$	No change
Stress-energy tensor	$T_{\mu\nu}$	No change

B.3 Relativistic Limits

The energy-momentum relation is $E^2 = p^2 c^2 + m^2 c^4$. So in a relativistic limit, we can have:

$$\{E, m, \rho\} \ll \{p, T\}$$

which any one of the values in left-hand set is much smaller than any one of the values in the right-hand set.

C.1 Physical Constants

Quantity	Symbol	Value (SI)
Speed of light	c	$2.998 \times 10^8 \text{ m s}^{-1}$
Gravitational constant	G	$6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$
Hubble constant today	H_0	$67.8 \pm 0.77 \text{ km s}^{-1} \text{ Mpc}^{-1}$
Cosmological constant	Λ	$1.11 \times 10^{-52} \text{ m}^{-2}$
Cosmological constant	Λ	$2.036 \times 10^{-35} \text{ s}^{-2}$
Planck constant	$\hbar$	$1.055 \times 10^{-34} \text{ J s}$
Boltzmann constant	k_B	$1.381 \times 10^{-23} \text{ J K}^{-1}$
Fermi constant	G_F	$1.166 \times 10^{-5} \text{ GeV}^{-2}$
Fine-structure constant today	α	$1/137.036$

C.2 Conversion Factors

Astronomical Units

Conversion	Value
1 astronomical unit (AU)	$1.496 \times 10^{11} \text{ m}$
1 light-year (ly)	$9.461 \times 10^{15} \text{ m}$
1 parsec (pc)	$3.086 \times 10^{16} \text{ m}$
1 parsec (pc)	3.262 ly
1 year (yr)	$3.156 \times 10^7 \text{ s}$
1 Solar mass ($M_\odot$)	$1.989 \times 10^{30} \text{ kg}$
1 solar radius ($R_\odot$)	$6.957 \times 10^8 \text{ m}$
1 Earth mass ($M_\oplus$)	$5.972 \times 10^{24} \text{ kg}$
1 Earth radius ($R_\oplus$)	$6.371 \times 10^6 \text{ m}$
CMB temperature (T_{CMB})	2.725 K
CνB temperature ($T_{\text{CνB}}$)	1.95 K

Energy Units

Conversion	Value
1 eV	$1.602 \times 10^{-19} \text{ J}$
1 electron mass (m_e)	$0.511 \text{ MeV}/c^2$
1 electron mass (m_e)	$9.109 \times 10^{-31} \text{ kg}$
1 proton mass (m_p)	$938.3 \text{ MeV}/c^2$
1 proton mass (m_p)	$1.673 \times 10^{-27} \text{ kg}$
1 neutron mass (m_n)	$939.6 \text{ MeV}/c^2$
1 neutron mass (m_n)	$1.675 \times 10^{-27} \text{ kg}$

C.3 SI Unit Prefixes

Prefix	Name	Factor
10^{-12}	pico (p)	10^{-12}
10^{-9}	nano (n)	10^{-9}
10^{-6}	micro (μ)	10^{-6}
10^{-3}	milli (m)	10^{-3}
10^3	kilo (k)	10^3
10^6	mega (M)	10^6
10^9	giga (G)	10^9
10^{12}	tera (T)	10^{12}

C.4 Main events in universe

Event	Time t	Redshift z	Temperature T (eV)
Planck epoch	10^{-43} s	—	—
Grand unification	10^{-36} s	—	—
Inflation	10^{-34} s (?)	—	—
Baryogenesis	?	?	?
EW phase transition	2×10^{-11} s	10^{15}	10^{11}
QCD phase transition	2×10^{-5} s	10^{12}	1.5×10^8
Dark matter freeze-out	?	?	?
Neutrino decoupling	1 s	6×10^9	10^6
Electron-positron annihilation	6 s	2×10^9	5×10^5
Big Bang nucleosynthesis	3 min	4×10^8	10^5
Matter-radiation equality	6×10^4 yr	3400	0.75
Recombination	$(2.6\text{--}3.8) \times 10^5$ yr	1100–1400	0.26–0.33
Photon decoupling	3.8×10^5 yr	1000–1200	0.23–0.28
Reionization	$(1\text{--}4) \times 10^8$ yr	11–30	$(2.6\text{--}7.0) \times 10^{-3}$
Dark energy-matter equality	9×10^9 yr	0.4	3.3×10^{-4}
Present	1.38×10^{10} yr	0	2.4×10^{-4}

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